

⚡ Problem Statement ID: SIH2025-EDTECH-03

EduMitra AI (एड्यूमित्र AI)

Centralized Multilingual AI Student Assistance & College Recommendation Platform for Rajasthan Technical Education

PROBLEM STATEMENT

Rajasthan Technical Admission Guidance

TEAM NAME

Endeavours

TARGET PORTALS

REAP & DTE Rajasthan

PRIMARY LANGUAGES

Hindi, English & Voice

🎯 Core Solution Innovation

1

Multilingual Voice & Text AI Chatbot

Instant natural language answers in Hindi & English for rural & urban students.

2

REAP Rank & Cutoff Predictor Engine

Data-driven college recommendations based on 5-year historical REAP/DTE cutoffs.

3

Automated Scholarship & Fee Matcher

Maps CM Higher Education & Govt welfare schemes to student profiles.

4

80% Workload Reduction for Colleges

Eliminates repetitive calls and counter visits for admission staff.

Multilingual Conversational AI

Voice & text interface powered by Bhashini AI supporting Hindi, English, and Rajasthani dialects to resolve queries on eligibility, hostels, fees, and cutoffs.

Smart REAP & DTE Rank Predictor

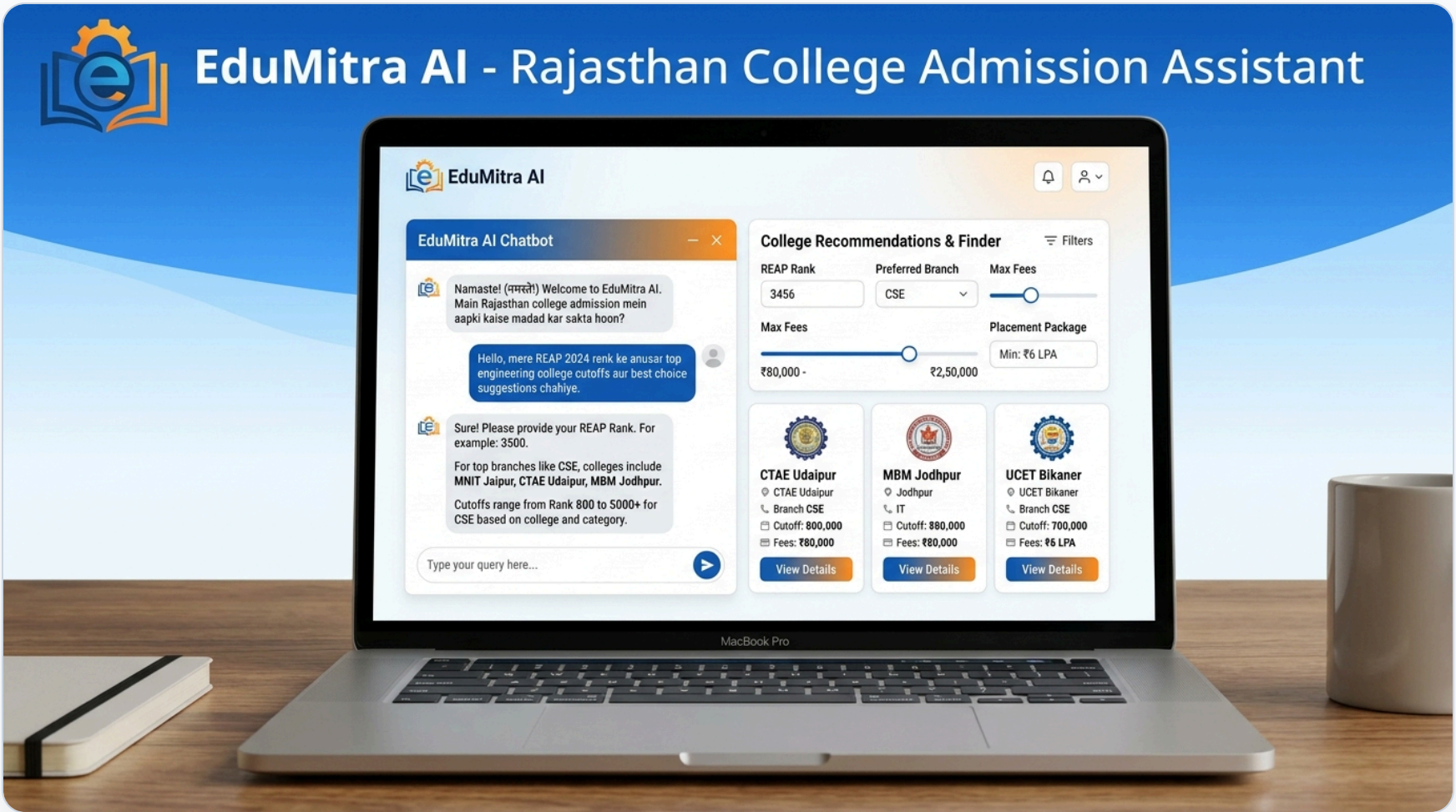
Algorithms analyze 5-year historical REAP percentile cutoffs to suggest optimal engineering & polytechnic colleges based on marks, rank, category, and preferred location.

Side-by-Side Institution Evaluator

Compare up to 4 colleges simultaneously across NAAC accreditation, median placement CTC, faculty ratio, hostel amenities, and total fee breakdown.

Scholarship & Financial Aid Matcher

Automatically identifies eligible state government scholarships (e.g., CM Higher Education Scheme, Post-Matric) to reduce financial barriers for rural students.



END-TO-END STUDENT WORKFLOW

REAL-TIME RAG FLOW

STEP 1

Student Query

Voice/Text in Hindi

STEP 2

AI Processing

RAG + REAP Vectors

STEP 3

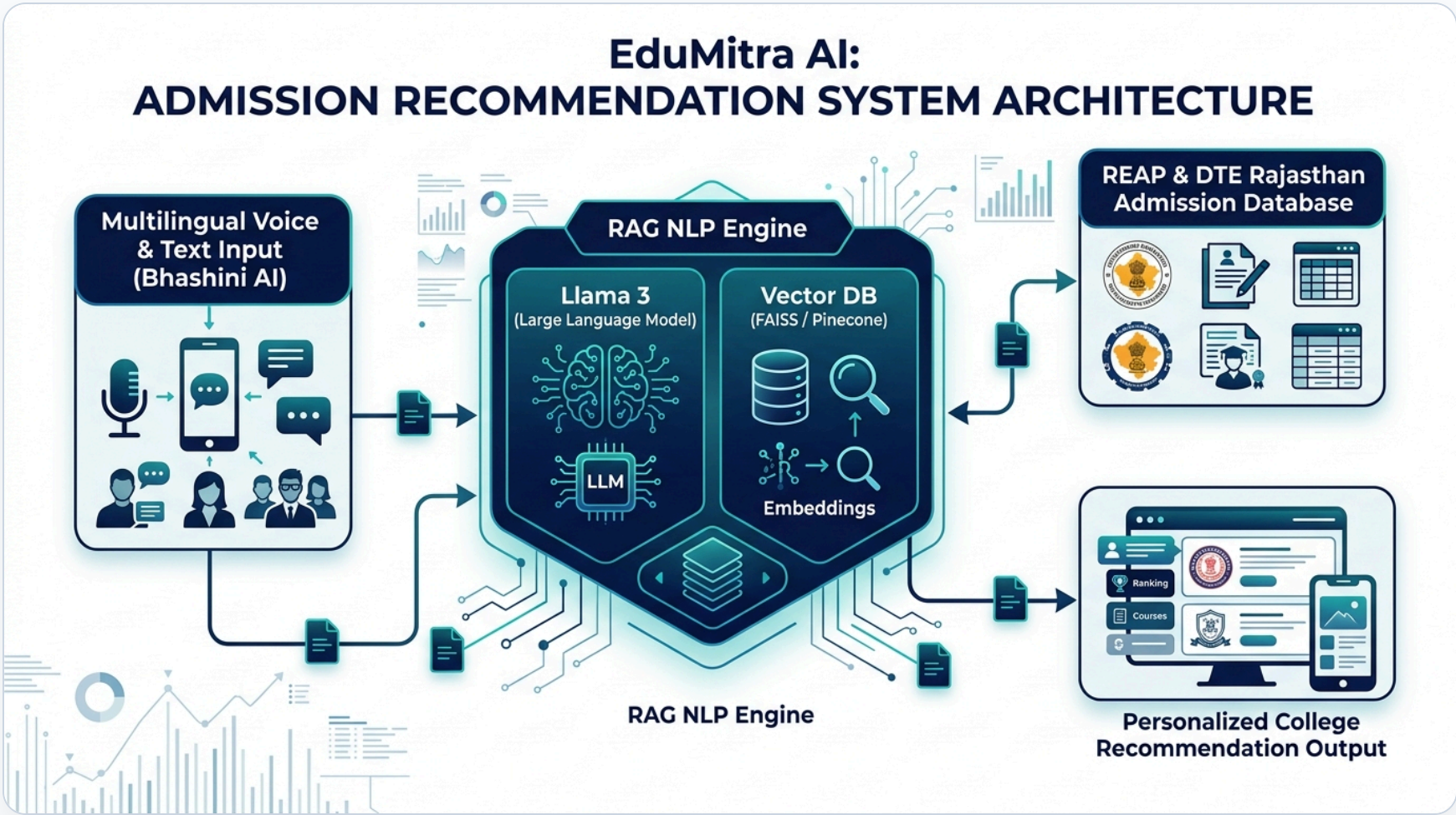
Match Output

Rank-Cutoff Matrix

STEP 4

Direct Action

Apply REAP Form



Complete Tech Stack Credentials

- React.js (Frontend PWA)
- Python FastAPI
- Node.js Express
- Llama-3 8B LLM
- ChromaDB Vector Store
- PostgreSQL
- Redis Caching
- Bhashini Voice STT/TTS
- Docker & K8s

- 01

User Access & Multilingual Layer
Responsive React.js PWA + WhatsApp API integration. Integrates **Bhashini Indic API** for real-time speech recognition (STT) and voice synthesis (TTS) in Hindi & English.
- 02

AI & RAG Intelligence Engine
Python FastAPI microservice utilizing **LangChain RAG pipeline** and **Llama-3 8B**. Queries vector embeddings of official REAP seat matrices and DTE handbooks stored in **ChromaDB**.
- 03

Relational Data & Cutoff Engine
PostgreSQL manages structured datasets for 150+ Rajasthan engineering & polytechnic colleges, previous year cutoffs, fees, placement stats, and hostel availability.
- 04

Zero Hallucination & Citation Engine
Strict RAG prompt grounding ensures AI responses are 100% cited from official DTE Rajasthan PDFs, eliminating misinformation in critical admission decisions.



High Technical Feasibility

Built using open-source, lightweight Llama 3 LLM and ChromaDB vector store. Can be self-hosted on cost-effective cloud infrastructure without heavy per-query API licensing costs.



Low-Bandwidth Rural Practicality

Optimized Progressive Web App (PWA) under 2MB payload size + WhatsApp Bot fallback ensures seamless operation across 2G/3G networks in remote districts of Rajasthan.



DTE Rajasthan Data Ingestion

Automated ETL data ingestion pipeline continuously updates seat matrices, fee changes, and REAP round-wise cutoffs directly from official government notifications.

⚠️ Risk Analysis & Mitigation Matrix

Potential Challenge / Risk	Impact	Mitigation Strategy
Dynamic Yearly REAP Seat Matrix Updates	MEDIUM	Automated Web Scraper + DTE Admin Verification Portal for fast dataset re-indexing.
Low Digital Literacy in Rural Parents	HIGH	Voice-first Hindi Q&A interface via Bhashini Speech API + WhatsApp bot integration.
AI Hallucination in Admission Advice	CRITICAL	Strict RAG grounding over DTE rulebooks with mandatory source URL citations.
Server Traffic Spikes During REAP Counselling	MEDIUM	Redis querv caching for popular FAQs + auto-scaling Docker microservices.

80%

Reduction in Manual Admission Inquiry Calls

< 2 Sec

Instant AI Response Time for Complex Queries

33

Rajasthan Districts Covered Seamlessly

100%

Free & Transparent Access for Students

 Beneficiary Impact Breakdown

- For Students & Parents:** Instant 24/7 clarity on cutoffs & fees without expensive travel or reliance on misleading admission agents.
- For Technical Colleges:** Offloads thousands of repetitive phone calls/emails during REAP admissions, freeing staff for campus operations.
- For DTE & Govt of Rajasthan:** Digitizes counselling guidance, promotes equitable access to rural students, and boosts scholarship enrollment.

 Scalability & Sustainability Model

- Direct DTE Rajasthan Deployment:** White-label integration into official `dte.rajasthan.gov.in` and REAP portal widgets.
- Pan-India State Expansion:** Reusable SaaS architecture scalable to other state technical admission boards (MHT-CET, WBJEE, UPTAC).
- Domain Expansion:** Extendable to ITI, Polytechnic diploma, Pharmacy, and Medical (NEET) counselling datasets.



Department of Technical Education (DTE), Rajasthan

Official notifications, polytechnic admission guidelines, fee structures, and institute affiliations. URL: dte.rajasthan.gov.in



Rajasthan Engineering Admission Process (REAP)

Official REAP seat matrix archives, round-wise cutoffs (2020–2024), and JEE Main / XII mark conversion rules. URL: reap2024.com



Bhashini — National Language Translation Mission

Open APIs for Indic language speech recognition (ASR) and text translation under Ministry of Electronics & IT (MeitY). URL: bhashini.gov.in



IEEE & ACM AI Literature References

Research paper: *"Retrieval-Augmented Generation for Public Governance Q&A Systems"* (IEEE Access 2024) & AI in Higher Education (ACM Digital Library).